



Feature

DILIRank 2.0: An updated and expanded database for drug-induced liver injury risk based on FDA labeling and a literature review

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Drug-induced liver injury (DILI) is of great concern in drug development and public health. DILIRank 1.0, a widely used public dataset that ranks FDA-approved drugs by their potential to cause DILI, has significantly enabled the development of new methods for improved DILI assessment. Here, we introduce DILIRank 2.0, an essential update of DILIRank 1.0, to capture new non-biologics drug and liver-related adverse event data generated since its inception 15 years ago. Using DILIRank 2.0, we also observe changes in the DILI profiles of approved drugs following the introduction of different FDA regulatory programs. DILIRank 2.0 is an up-to-date resource that offers opportunities for supporting DILI safety assessment, predictive model development, and evaluation of new approach methods (NAMs).

Keywords: drug-induced liver injury (DILI); DILIRank; hepatotoxicity; database; drug safety

Introduction

DILI emerged as a critical concern in healthcare during the late 20th century, largely because of the seminal work of Hyman Zimmerman.^(p1) Since then, research interest in DILI has expanded significantly. Investigations have included, but are not limited to, characterizing clinical cases of DILI, categorizing drugs based on their hepatotoxic potential,^(p2) identifying drug-related and host-related risk factors,^{(p3),(p4)} standardizing diagnostic criteria,^(p5) elucidating the underlying

mechanisms of hepatotoxicity,^{(p6),(p7)} and developing predictive models for DILI assessment.^{(p8),(p9)} These efforts aim to enhance the understanding of DILI pathogenesis, improve early detection and prevention strategies, and ultimately reduce its incidence in humans. Advancements in this field are expected to alleviate the healthcare burden associated with DILI and improve the success rates of drug development by reducing late-stage attrition caused by hepatotoxicity.

Annotating drugs based on their DILI risk in humans is a critical step toward advancing the study of DILI, and especially in developing predictive models, which utilize existing knowledge of hepatotoxic potential to construct datasets for model development. Previously, we introduced a DILI benchmark dataset, a systematic classification scheme for DILI based on data extracted from the FDA-approved drug labels.^(p10) This classification was later refined by incorporating

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clinical case data, where DILI causality was verified through standardized assessment methods, such as the Roussel Uclaf Causality Assessment Method (RUCAM), or expert consensus.^(p11) The resulting DILIRank (version 1.0) dataset, comprising 1036 drugs, has had a significant impact, as evidenced by its extensive acceptance within the scientific community. DILIRank 1.0 has been recommended for DILI annotation use^{(p12),(p13)} and has served as a reference for DILI classification in numerous studies, including studies on *in silico* model development, assessment, and validation^{(p14),(p15),(p16),(p17),(p18),(p19)}, for identification of DILI risk factors^{(p20),(p21)}; and in the evaluation of *in vitro* assay predictive performance.^{(p9),(p22)}

A comprehensive review of DILI-specific drug databases highlighted DILIRank 1.0 as a valuable resource, with a notable limitation being the need for an update of new drug molecules approved by the FDA recently.^(p23) DILIRank 1.0 contains drugs approved by the FDA only up to 2009. Given that the FDA approves ~40 new drugs annually, ~400 drugs approved between 2010 and 2021 are not present in DILIRank 1.0. Additionally, new safety concerns have emerged for some drugs that were annotated 15 years ago. To ensure that the DILIRank dataset remains current and includes the most recent FDA-approved drugs, a new update of DILIRank dataset is required.

Here, we present DILIRank 2.0, a critically needed update to the DILIRank database.^(p11) DILIRank 2.0 incorporates revisions of the previous version based on DILI cases newly reported in the literature and annotating additional drugs approved by the FDA through 2021, in recognition of the need for time to discover and report safety concerns during post-marketing surveillance. In addition to updating the database, we also analyze the impact of historical regulatory policies on the DILI landscape among FDA-approved drugs.

DILIRank 2.0 Part I – update to DILIRank 1.0 drug annotations

Given the low incidence of DILI in humans and the current limitations of predictive models in accurately identifying hepatotoxicity, some toxic drugs might evade detection during preclinical testing and clinical trials. This challenge is partic-

ularly pronounced for drugs that induce liver injury in an idiosyncratic manner. Therefore sufficient time is required for post-marketing surveillance to allow for the development and assessment of clinical case reports confirming DILI causality.

Here, we reassessed the annotations from DILIRank 1.0 by incorporating newly reported DILI cases from the literature, focusing on drugs previously classified as ^vNo-DILI-Concern or Ambiguous-DILI-Concern. In contrast, drugs previously categorized under the ^vLess- and ^vMost-DILI-Concern groups already met our verification criteria, and their DILI classification remained unchanged (Figure 1a).

To evaluate recent evidence of DILI causality, we conducted a literature search for peer-reviewed case reports using the keywords: [drug name] AND (liver injury OR DILI OR hepatotoxicity OR RUCAM OR CIOMS OR Naranjo OR causality OR induced). Reports were considered only if they were written in English and the cases were adjudicated using a causality assessment method (CAM) such as RUCAM. Similar to DILIRank 1.0, a drug was only 'verified' as causing DILI if the CAM score was categorized as 'definite' or 'probable' in the Naranjo scale; or 'highly probable', 'probable', or 'possible' in the Council for International Organizations of Medical Sciences (CIOMS)/RUCAM scales. A drug was deemed 'not verified' as a cause of DILI if no case reports were found, if the available case reports did not apply a CAM score, or if the CAM score was assigned to a combination of two or more drugs. Table S1 in the supplementary material online lists these drugs, which had DILI-related case reports but did not meet our verification criteria, along with the references to the relevant articles.

Of the 312 drugs classified as ^vNo-DILI-Concern in DILIRank 1.0, DILI case reports were found for 43 drugs. Among these, 18 drugs met our verification criteria for causality and were reclassified as ^vLess-DILI-Concern in DILIRank 2.0, whereas the remaining 25 drugs retained their ^vNo-DILI-Concern classification. Similarly, DILI case reports were found for 76 out of 254 drugs previously categorized as Ambiguous-DILI-Concern. Of these, 31 drugs were newly verified to cause DILI. Based on the drug labeling information and severity of DILI described in their

FDA-approved drug labels, 27 drugs were reclassified as ^vLess-DILI-Concern, whereas 4 drugs were reclassified as ^vMost-DILI-Concern. The DILI classification for the remaining 45 drugs was left unchanged (Figure 1a).

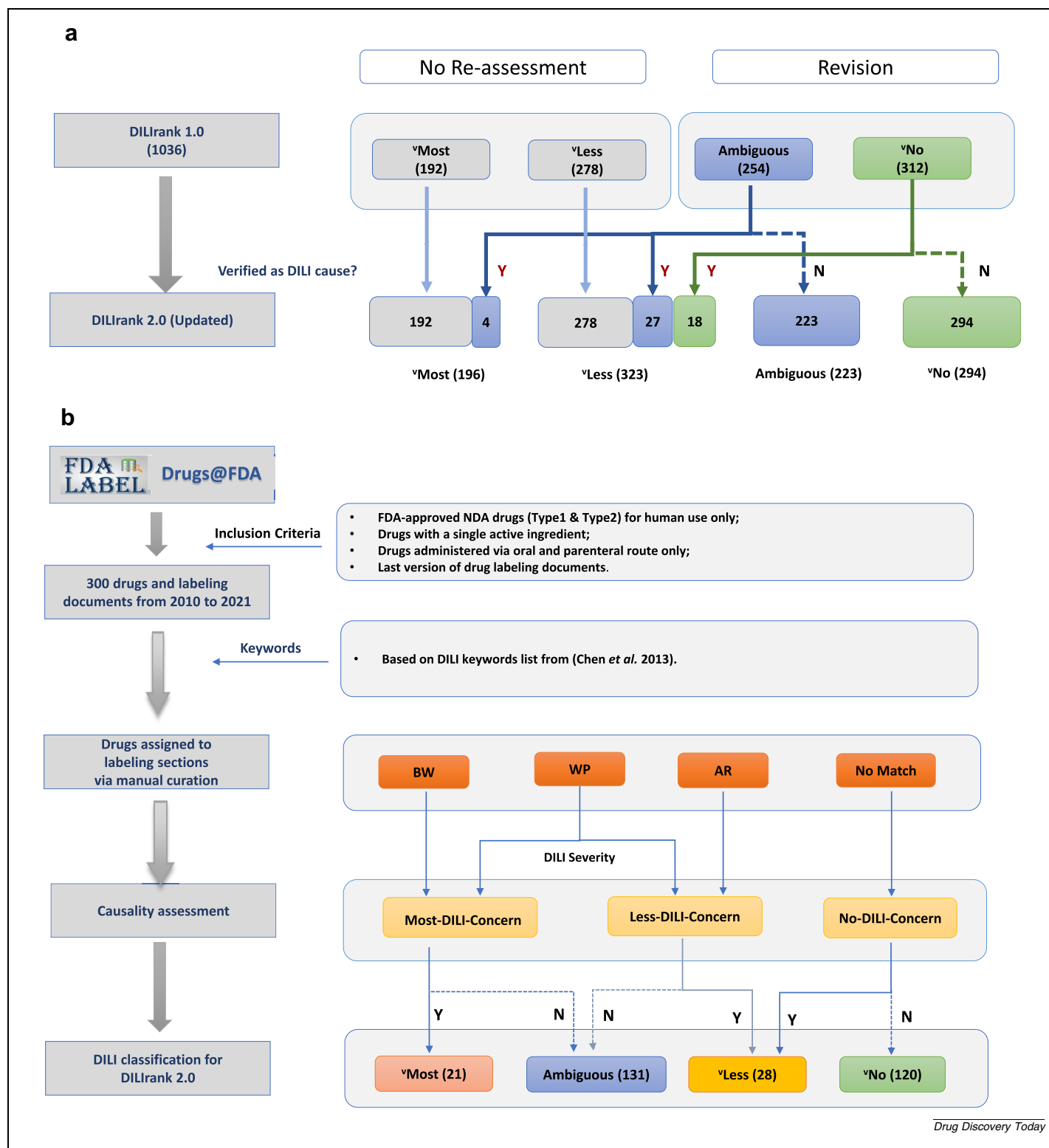
DILIRank 2.0 Part II – addition of 300 new drugs

Because DILIRank 1.0 contains drugs approved up to 2009, there is a need to thoroughly examine and annotate new drugs that have been approved since then. Thus, in addition to updating DILIRank 1.0, we also expanded the database to classify drugs approved by the FDA from 2010 to 2021 by applying our former schema, as used for DILIRank 1.0 and described in Chen *et al.*^(p11) (Figure 1b). The selection process involved filtering drugs based on the following criteria:

- (i) type of new drug application (NDA) for FDA-approved drugs: type 1 NDA (new molecular entity) or type 2 NDA (new active ingredient that is not a new molecular entity);
- (ii) drugs containing a single active ingredient only;
- (iii) drugs intended for human use only;
- (iv) drugs administered exclusively via oral or parenteral routes; and
- (v) drugs possessing an active FDA drug label.

For drugs meeting these criteria, the most recent version of the drug label was retrieved from the FDALabel database^(p24) (version 2.7.1, 16 February 2023) and used to assign DILI severity and establish the initial DILI classification, following the previously described methodology.^(p10) Finally, the assigned DILI classification was verified using the same approach applied in the re-evaluation of DILI annotation, as outlined in the preceding section. Drugs for which hepatotoxicity concerns were described in boxed warning sections of the FDALabel document were considered verified as causing DILI, as previously done.

A total of 300 new FDA-approved drugs were incorporated into DILIRank 2.0. Among these, 49 drugs were verified as causing DILI, with 28 drugs classified as ^vLess-DILI-Concern and 21 as ^vMost-DILI-Concern. Additionally, 120 drugs were cat-

**FIGURE 1**

Workflows for the annotation of DILrank 2.0. Based on available literature, a drug was either verified to cause DILI (Y) or not (N). DILI is grouped as ^YMost-DILI-Concern (^YMost), Ambiguous DILI-Concern (Ambiguous), ^YLess-DILI-Concern (^YLess), or ^YNo-DILI-Concern (^YNo). (a) Revision of the DILI annotation of 49 drugs in DILrank 1.0. (b) DILI classification and annotation for 300 drugs approved by the FDA from 2010 to 2021. Drug labeling sections containing DILI information matching our keywords were boxed warning (BW), warnings and precautions (WP), and adverse reactions (AR); 'No Match' was used if no information was in these sections of the labeling document (See Chen *et al.* 2016^(p11)).

egorized as ^vNo-DILI-Concern, whereas the remaining 131 drugs were designated as Ambiguous-DILI-Concern.

Analysis of the DILrank 2.0 landscape

Following revision and expansion, DILrank 2.0 contains a total of 1336 drugs, categorized as follows: 414 ^vNo-DILI-Concern, 354 Ambiguous DILI-Concern, 351 ^vLess-DILI-Concern, and 217 ^vMost-DILI-Concern drugs. With the addition of 300 drugs to DILrank 1.0, the total number of drugs in DILrank 2.0 increased by 29% (Table 1). Despite this, it is interesting to note that in DILrank 2.0, the number of drugs classified as ^vMost-DILI-Concern decreased by 3% compared with DILrank 1.0. Meanwhile, the ^vNo- and Ambiguous-DILI-Concern drugs (drugs that were not verified to cause DILI) increased in proportion in DILrank 2.0.

No biological product approved by the FDA between 2010 and 2021 is included in DILrank 2.0, as these require approval through a Biologics License Application (BLA), a process distinct from the NDA pathway. Additionally, for drugs approved under a type 2 NDA, different salt forms of the same base drug were included as separate entries in DILrank 2.0 (e.g. amifampridine and amifampridine phosphate). This distinction acknowledges that different salt forms can exhibit unique pharmacokinetic properties, including hepatic exposure, which could be a critical factor in DILI. Furthermore, this disparity was also reflected in the drug labels. For instance, the drug label for amifampridine does not contain any hepatotoxicity-related information, leading to its classification as ^vNo-DILI-Concern in DILrank 2.0, with a DILI severity of 0. In contrast, the adverse reactions section of the drug

label for amifampridine phosphate reports elevations in liver enzymes observed in clinical trials, resulting in its classification as Ambiguous-DILI-Concern, with a DILI severity of 3. At the time of this study, neither drug met the verification criteria for confirmed DILI causality.

To assess the chemical space distribution of drugs in both versions of DILrank, the MACCS molecular fingerprint was computed for 1265 drugs whose chemical structures were retrieved from the PubChem database using the RDKit module in Python.^{(p25),(p26)} The first three principal components, following a principal component analysis of the molecular fingerprints, account for 23% of the total variation (see Figure S1 in the supplementary material online). Thus, a visualization of the chemical space is given using a t-distributed stochastic neighbor embedding (SNE) model trained on principal components of the molecular fingerprints (Figure 2a). The chemical space coverage of drugs in DILrank 2.0 reveals a slight expansion from that of DILrank 1.0, suggesting that there is no significant difference in the chemical space occupied by the newly added drugs in DILrank 2.0 compared with drugs in the DILrank 1.0.

In addition, we identified five new pairs of drugs with ‘activity cliffs’, where at least one drug of the pair is a new addition to DILrank 2.0 (Table 2). These drug pairs have similar chemical structures and therapeutic profiles but have significantly distinct hepatotoxicity in humans, supplementing a previously published list.^(p8) The therapeutic profiles of drugs were determined using the WHO Anatomical Therapeutic Chemical (ATC) classification system. The system provides multilevel classification for drugs into

anatomical/pharmacological, therapeutic, and chemical (ATC) groups, and is endorsed by the WHO as an international standard for drug utilization monitoring and research.^{(p27),(p28)} In this study, the drug pairs were selected using the following criteria:

- (i) Chemical similarity: the Tanimoto similarity coefficient, calculated based on the (Molecular ACCess System) (MACCS) molecular fingerprint, was at least 0.6.
- (ii) Therapeutic similarity: the drug pair belongs to the same level 4 ATC subgroup of the WHO ATC classification system, but excluding nonspecific subgroups, which start with the word ‘Other’.
- (iii) Difference in hepatotoxicity: one drug of the pair was classified as ^vNo-DILI-Concern, and the other as ^vMost-DILI-Concern in DILrank 2.0.

DILrank 2.0 contains new drugs added to all 14 first level WHO ATC classification categories. Some drugs have multiple ATC codes and are classified in all relevant therapeutic categories. The relative sizes of most ATC categories are comparable between DILrank 1.0 and DILrank 2.0, with one main exception, class L (antineoplastic and immunomodulating agents), which increased from 144 drugs (12%) to 244 drugs (16%) (Figure 2b). The substantial increase reflects the approval by the FDA of 100 cancer-related drugs over the decade.

Based on the second level of the WHO ATC classification system (ATC level 2), which describes the pharmacological or therapeutic action of a drug, 78 antineoplastic agents (L01) and 17 immunosuppressants (L04) were added in DILrank 2.0. The names of all ATC level 2 subgroups for the DILrank 2.0 drugs are listed in Table S2 in the supplementary material online. Most of the newly added drugs that were verified as causing DILI (classified as ^vLess- or ^vMost-DILI-Concern) belong to these two classes. Distribution of the drugs in DILrank 2.0 by their therapeutic action and DILI classification reveals that the L01 subgroup has the highest number of DILI-positive (^vLess- and ^vMost-DILI-Concern) drugs (Figure 3). Subgroups enriched in DILI-positive drugs are marked with an

TABLE 1

Differences between DILrank 1.0 and DILrank 2.0.

	DILrank 1.0 ^a	DILrank 2.0 ^a	Increase ^b
Total number of drugs	1036	1336	300 (29%)
Number of drugs per DILI group (%)			
^v No-DILI-Concern	312 (30%)	414 (31%)	102 (33%)
Ambiguous-DILI-Concern	254 (25%)	354 (26%)	100 (39%)
^v Less-DILI-Concern	278 (27%)	351 (26%)	73 (26%)
^v Most-DILI-Concern	192 (19%)	217 (16%)	25 (13%)

^a Percentage computed based on all groups in database version.

^b Percentage computed based on each group increase across both database versions.

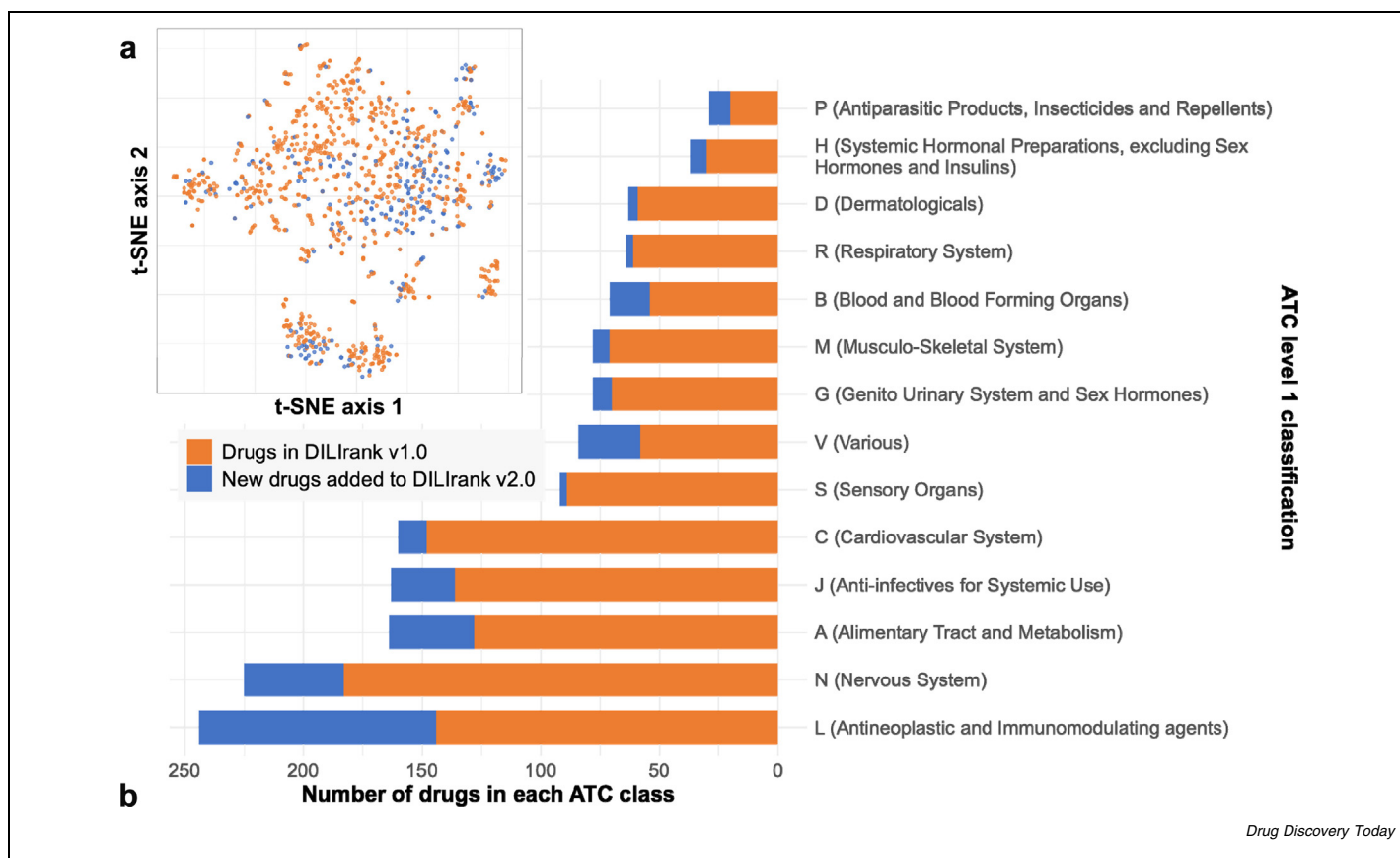


FIGURE 2 Differences between DILIRank 1.0 and 2.0: **(a)** Chemical space coverage of drugs, projected on a t-SNE plot; **(b)** Distribution of drugs by the main anatomical ATC classes in DILIRank (drugs with no ATC code are excluded from the plot).

asterisk in Figure 3. A class is considered enriched if the enrichment factor (EF) is greater than or equal to 2. EF is calculated as $(P_i/N_i)/(\sum P_i/\sum N_i)$, where P_i and N_i represent the number of DILI-positive and DILI-negative drugs ($^{\vee}$ No-DILI-Concern) in each class, respectively.

Over the past decade, there has been significant scientific interest in protein kinase inhibitor-induced hepatotoxicity, with several DILI cases reported.^{(p29), (p30), (p31)} DILIRank 2.0 also reflects this trend, as only one of the newly added protein kinase inhibitors is classified as $^{\vee}$ No-DILI-Concern, whereas 18 met our verification criteria and were classified as $^{\vee}$ Less- or $^{\vee}$ Most-DILI-Concern. Most of the drugs (36) are classified as Ambiguous DILI-Concern, because despite the hepatotoxicity warning in their drug labels, the case reports for these drugs did not meet our verification criteria.

DILI profile changes following introduction of FDA regulatory programs

The regulatory policies of the FDA are crucial in ensuring drug safety by enforcing rigorous premarket evaluations, post-marketing surveillance, and risk mitigation strategies to detect and manage adverse drug reactions, including DILI. DILIRank 2.0 contains drugs approved by the FDA from 1939 to 2021, offering an unprecedented opportunity to assess trends in the drug DILI risk profile following the introduction of FDA programs. In this study, we specifically examined the influence of three key regulatory programs on the DILI liability of approved drugs: (i) FAERS (FDA Adverse Event Reporting System), established in 1969; (ii) expedited approval programs, introduced in 1992; and (iii) the DILI-specific guidance, issued in 2009. The analysis is based on 1265 drugs in DILIRank 2.0 for which we could identify the year of FDA drug approval.

The first regulatory milestone we examined was the establishment of FAERS in 1969, which introduced a mechanism for the spontaneous reporting of drug-related adverse events and medication errors.^(p32) Before FAERS, a series of high-profile drug-induced adverse events heightened public concern over drug safety. One of the most notable cases was thalidomide, which was found in 1962 to cause severe congenital abnormalities in newborns across Europe.^(p33) In response, the US Congress passed the Kefauver–Harris Drug Amendments to the Federal Food, Drug, and Cosmetic (FDC) Act, strengthening regulations to ensure both drug efficacy and safety.^(p34) As illustrated in Figure 4, DILIRank 2.0 contains 218 drugs approved between 1939 and 1968 (pre-FAERS) and 309 drugs approved between 1969 and 1991 (post-FAERS). Over this period, the proportion of $^{\vee}$ Most- and $^{\vee}$ No-DILI-concern drugs decreased from 14% to

TABLE 2

Drug pairs with DILI-related activity cliffs.

S/N	LTKBID	Drug name	DILI-Concern	Tanimoto similarity	Therapeutic category
1	LT02444 LT01600	Enzalutamide Nilutamide	^v No- ^v Most-	0.63	Anti-androgens
2	LT03745 LT00034	Rifamycin sodium Rifampin	^v No- ^v Most-	0.64	Antibiotics
3	LT03753 LT01222	Sarecycline hydrochloride Tigecycline	^v No- ^v Most-	0.77	Tetracyclines
4	LT03753 LT00416	Sarecycline hydrochloride Minocycline hydrochloride	^v No- ^v Most-	0.79	Tetracyclines
5	LT03563 LT00380	Voclosporin Cyclosporine	^v No- ^v Most-	0.98	Calcineurin inhibitors

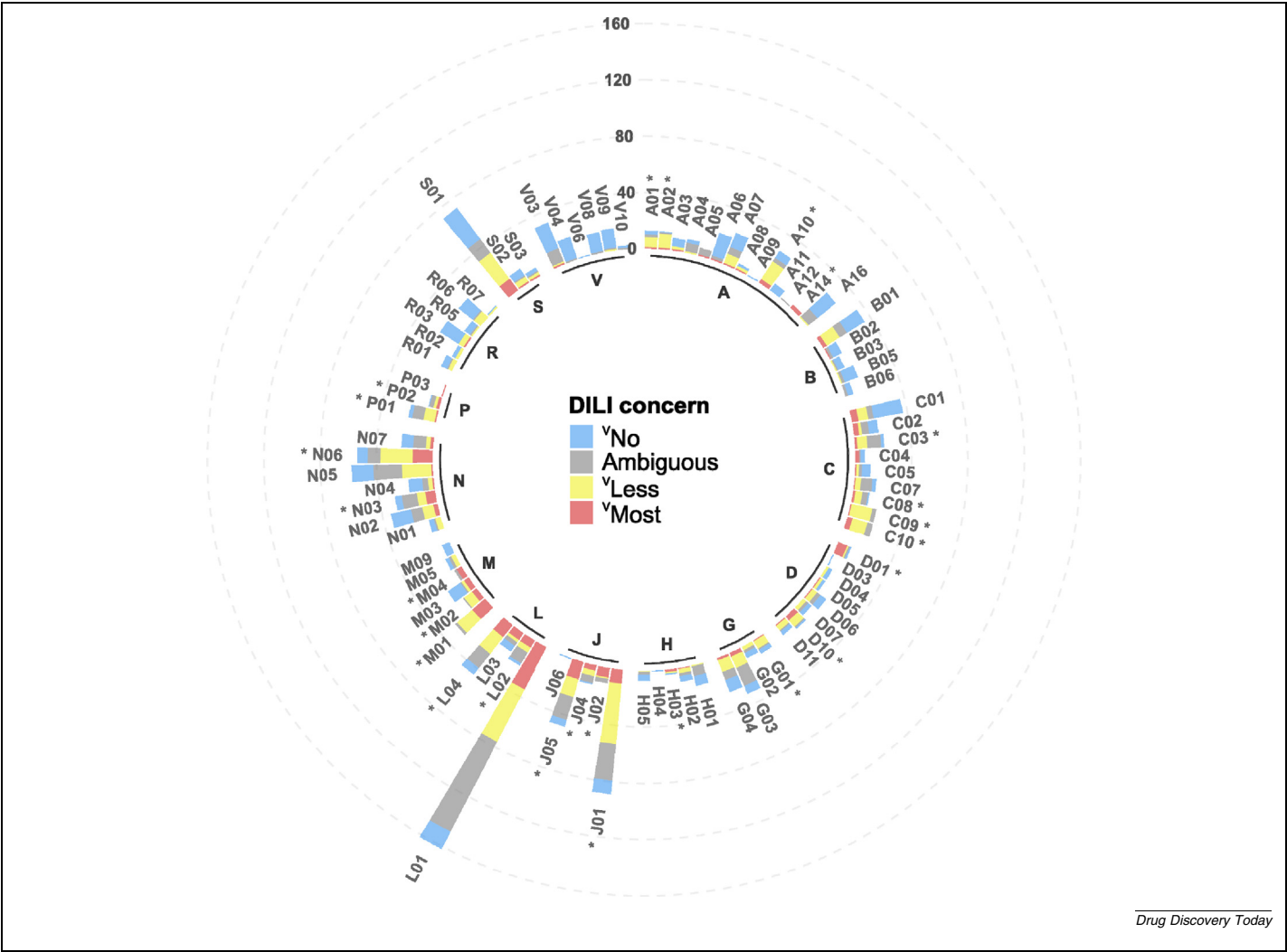
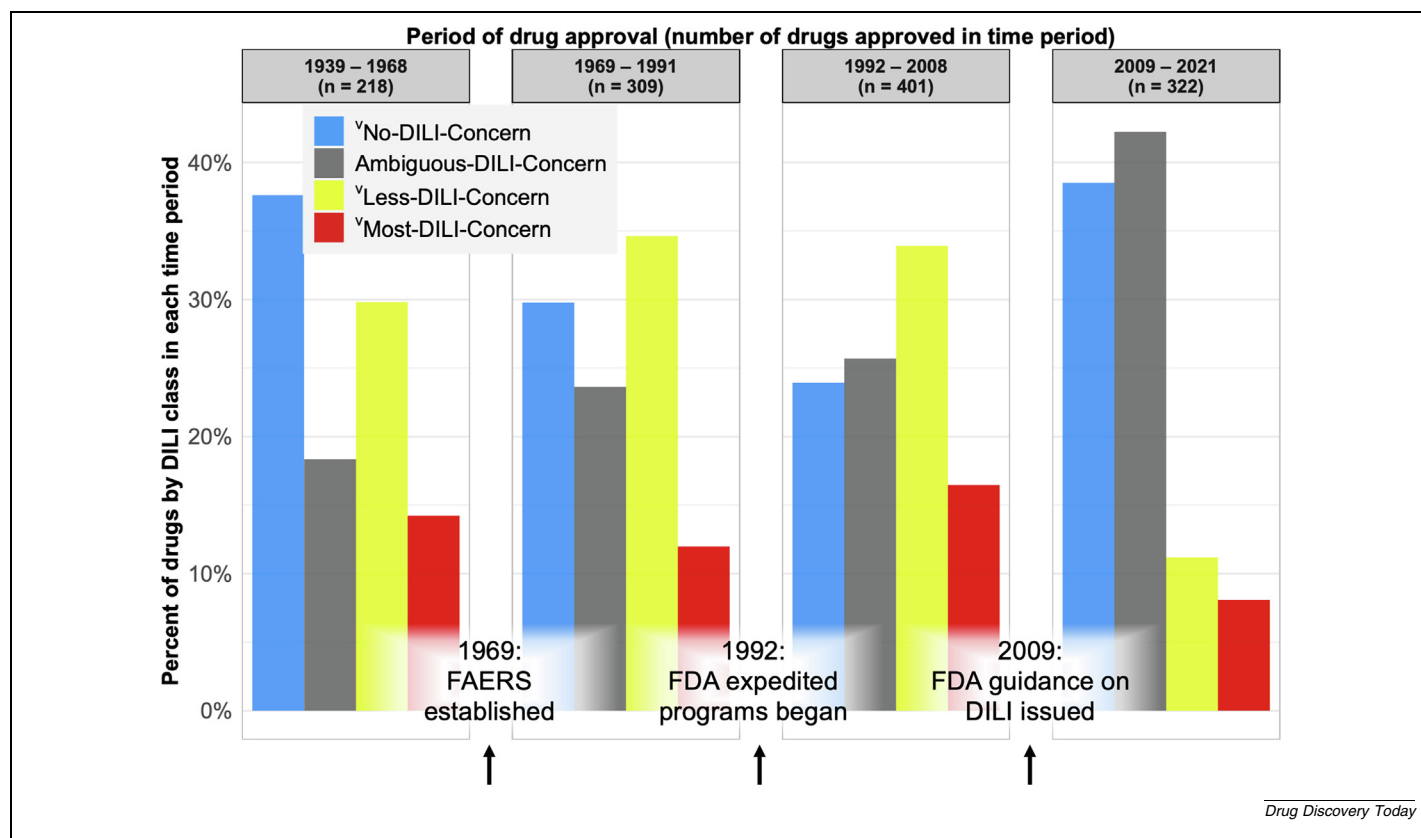


FIGURE 3

Distribution of DILIRank 2.0 by the WHO ATC level 2 class (pharmacological or therapeutic subgroup) and DILI classification. Each bar height represents the number of drugs in each class. The names of all ATC level 2 subgroups are listed in Table S2.

**FIGURE 4**

Drugs in DILIRank 2.0 by DILI classification and year of approval.

12% and from 38% to 30%, respectively, during the pre-FAERS to the post-FAERS era. Meanwhile, the proportion of 'Less- and Ambiguous-DILI-Concern drugs slightly increased. Based on the changes in all four groups during this period, it is difficult to ascertain the specific influence of the FAERS system on the DILI risk profile of FDA approved drugs. Nevertheless, it is worth noting that since its major revision in 1997, the FAERS database has become the largest repository of such reports, and its data have been mined several times to establish links between drugs and their reported adverse events.^(p35)

In the late 1980s, the FDA officially stated its interest in 'expediting the availability of promising new therapies' to accelerate the development and approval of drugs addressing serious unmet medical needs.^(p36) The FDA employed four programs, initiated at different times to expedite its review and approval process:

Priority Review (1992), Accelerated Approval (1992), Fast Track (1998), and Breakthrough Therapy Approval (2012).^(p37) According to multiple investigations into drugs approved by the FDA between 1987 and 2023, Fast Track and Priority Review are the two most used of the four FDA expedited programs.^{(p37),(p38),(p39),(p40),(p41),(p42)} However, concerns have been raised that expedited approval processes might be associated with a higher likelihood of post-marketing drug safety issues.^{(p43),(p44)} As shown in Figure 4, compared with drugs approved between 1969 and 1991, we observed a higher proportion of 'Most-DILI-Concern drugs among those approved during 1992–2008 (16%). Simultaneously, the proportion of 'No-DILI-Concern drugs declined from 30% (1969–1991) to 24% (1992–2008). These findings suggest that drugs approved through expedited programs might carry a higher risk of DILI, empha-

sizing the need for enhanced post-marketing surveillance to monitor and mitigate potential safety concerns. It is also worth noting that expedited programs are generally employed for drugs to be used in cases where there is a willingness to 'accept greater risks and side effects from treatment of life-threatening and severely debilitating disease'.^(p45) In line with this, DILIRank 2.0 shows that the majority of the drugs that were approved through an FDA expedited program and are classified as 'Most-DILI-Concern (see Table S3 in supplementary material online) are antineoplastic agents (ATC code: L01) and antivirals for systemic use (ATC code: J05).

In 2009, the FDA issued guidance on conducting premarketing clinical evaluation of DILI.^(p46) Notably, the liver safety profile of newly approved drugs has significantly improved in subsequent years (2009–2021). During this period, the

proportion of ^vNo-DILI-Concern drugs increased to 39%; whereas the proportions of DILI-positive drugs and ^vLess- and ^vMost-DILI-Concern drugs dropped to all-time lows of 11% and 8%, respectively. These findings suggest that following the introduction of the DILI guidance there has been a positive trend in mitigating DILI risk associated with FDA-approved drugs. However, it is important to note that this period also records the highest proportion of Ambiguous DILI-Concern drugs (42%). This might reflect raised awareness and caution in reporting hepatotoxicity concerns during clinical trials, and in drug labels. Because post-marketing surveillance requires sufficient time to detect late-emerging safety signals, there will be a continuous need to assess and update the DILI risk of these drugs in the future.

Concluding remarks

DILIRank 2.0 is an updated and expanded dataset with DILI classification for 1336 drugs approved by the FDA from 1939 to 2021 under four DILI-concern categories. DILIRank 2.0 offers a valuable resource to enable further studies on DILI, particularly those using NAMs including arti-

cial intelligence (AI)-based approaches. Based on this update, we observed changes in liver safety profiles of drugs approved over a period of eight decades following the introduction of FDA regulatory programs. As drug development efforts advance, there will be a need to continue annotating new drugs for their potential to cause DILI and to reassess the annotations of former drugs as new evidence emerges.

Declarations of interest

There are no conflicts of interest to declare.

CRediT authorship contribution statement

AyoOluwa O. Olubamiwa: Data curation, Formal analysis, Methodology, Writing – original draft; **Yanyan Qu:** Data curation, Formal analysis, Writing – review and editing; **Skylar Connor:** Formal analysis, Writing – review and editing; **Weida Tong:** Conceptualization, Formal analysis, Writing – review and editing; **Dongying Li:** Data curation, Formal analysis, Project administration, Writing – review and editing; **Minjun Chen:** Conceptualization, Data curation, Formal analysis, Methodology,

Writing – original draft, Writing – review and editing.

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Appendix A. Supplementary material

Supplementary material to this article can be found online at <https://doi.org/10.1016/j.drudis.2025.104485>.

Data availability

The DILIRank 2.0 dataset is available online at <https://www.fda.gov/science-research/liver-toxicity-knowledge-base-ltkb/drug-induced-liver-injury-rank-DILIRank-dataset>.

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